

Lecture 12 (153-248)

Bayesian Regularization

Recap: (Model & Ridge Regularization)

$$y_1, \dots, y_n$$

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2) + \beta_3(t-3) + \dots + \beta_{n-1}(t-(n-1)) + \varepsilon_t$$

$\varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$

$$y = X\beta + \varepsilon$$

$$X = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ 1 & 2 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n-1 & n-2 & \dots & 1 \end{bmatrix}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_{n-1} \end{pmatrix}$$

sm.OLS(y, X)

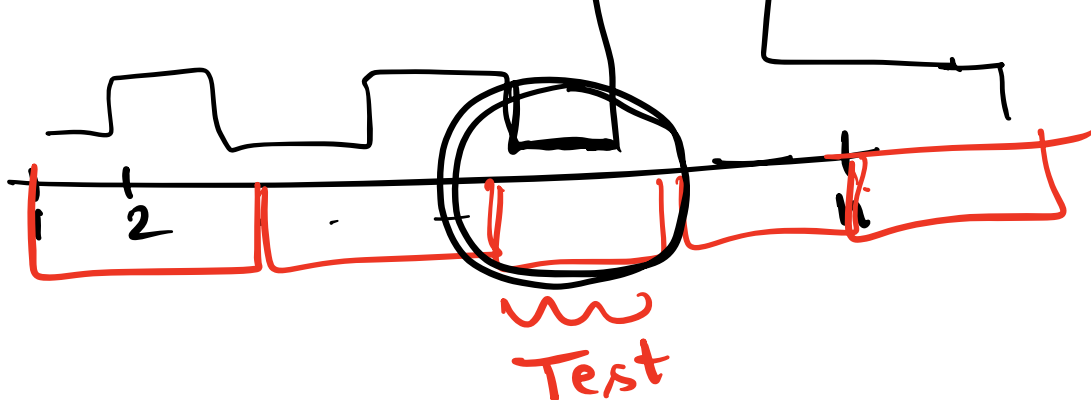
Regularization (RIDGE)

Minimize $\|y - X\beta\|^2 + \lambda(\beta_1^2 + \dots + \beta_{n-1}^2)$

CVXPY $\rightarrow$ for some $\lambda \geq 0$

CROSS-VALIDATION

$\hat{\beta}_{\text{ridge}}(\lambda)$



Differentiate
w.r.t β
(take the gradient)

$$\nabla_{\beta} \|y - X\beta\|^2 = 2X^T X \beta - 2X^T y$$

$$\nabla_{\beta} (\beta_2^2 + \dots + \beta_{n-1}^2)$$

$$= \begin{pmatrix} 0 \\ 0 \\ 2\beta_2 \\ 2\beta_3 \\ \vdots \\ 2\beta_{n-1} \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{pmatrix}$$

$$\begin{pmatrix} X^T X \beta = X^T y \\ \beta = (X^T X)^{-1} X^T y \end{pmatrix}$$

$$\nabla_{\beta} \left[\|y - X\beta\|^2 + \lambda (\beta_2^2 + \dots + \beta_{n-1}^2) \right]$$

$$= 2X^T X \beta - 2X^T y + 2\lambda \begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \vdots & \vdots \\ 0 & 0 & 1 \end{bmatrix} \beta$$

$$J = \begin{bmatrix} 0 & 0 & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = \text{diag}(0, 0, 1, \dots, 1)$$

$$\nabla_{\beta} \left(\|y - X\beta\|^2 + \lambda (\beta_2^2 + \dots + \beta_{n-1}^2) \right) \\ = 2 X^T X \beta - 2 X^T y + 2 \lambda J \beta = 0$$

$$(X^T X + \lambda J) \beta = X^T y$$

$$\Rightarrow \hat{\beta}_{\text{ridge}}(\lambda) = \underbrace{(X^T X + \lambda J)^{-1}}_{n \times n} X^T y$$

sm.OLS
 $(X^T X)^{-1} X^T y$

BAYESIAN REGULARIZATION for RIDGE

$$y = X\beta + \varepsilon$$

$$X = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ & 1 & 0 & \dots & 0 \\ & & 2 & \dots & 0 \\ & & & \ddots & \\ 1 & n-1 & n-2 & \dots & 1 \end{pmatrix}$$

(Practically,
NO CV here)

Recall: $\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^n \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2} \right) \rightarrow \text{Likelihood}$

Prior: $\beta_0, \beta_1, \dots, \beta_{n-1} \stackrel{iid}{\sim} \text{Unif}(-C, C)$
 $C \rightarrow \infty$

$\Rightarrow$ posterior:

$$\beta \mid \text{data} \sim N\left(\underbrace{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}}_{\text{least squares}}, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}\right)$$

Different Regularization

$\beta_0 \dots \beta_{n-1} \stackrel{iid}{\sim} N(0, C)$
 (SAME AS $\text{Unif}(-C, C)$
 when C is large)

$$\frac{1}{\sqrt{2\pi} \sqrt{C}} \exp\left(-\frac{\beta_j^2}{2C}\right)$$

$$= \frac{1}{\sqrt{2\pi} \sqrt{C}} \propto 1$$

$\beta_0 \dots \beta_{n-1} \stackrel{iid}{\sim} N(0, C)$

$\downarrow$

posterior $\beta \mid \text{data}$
 σ

$$\sim N\left(\left(\frac{\mathbf{X}^T \mathbf{X}}{\sigma^2} + \frac{\mathbf{I}}{C}\right)^{-1} \frac{\mathbf{X}^T \mathbf{y}}{\sigma^2}, \left(\frac{\mathbf{X}^T \mathbf{X}}{\sigma^2} + \frac{\mathbf{I}}{C}\right)^{-1}\right)$$

$\beta_0 \dots \beta_{n-1} \stackrel{iid}{\sim} \text{Unif}(-C, C)$

$$(\beta_j) \sim \frac{\mathbf{I}(-C < \beta_j < C)}{2C}$$

$$f_{\beta_j}(x) = 1$$

I : identity of order $n \times n$.

$$\beta | \text{data} \sim N\left(\left(\frac{X^T X}{\sigma^2} + \frac{I}{C}\right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + \frac{I}{C}\right)^{-1}\right)$$

$$\xrightarrow{C \rightarrow \infty} N\left(\left(\frac{X^T X}{\sigma^2}\right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2}\right)^{-1}\right)$$

$$= N(X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1}$$

LEAST SQUARES = $\underset{\sim}{\text{iid}} N(0, C)$
prior for $C \rightarrow \infty$

= $\underset{\sim}{\text{iid}} \text{Unif}(-C, C)$
prior for $C \rightarrow \infty$

New Prior $\beta_0 \dots \beta_{n-1}$

$$\beta_0, \beta_1 \sim N(0, C), \beta_2 \dots \beta_{n-1} \underset{\sim}{\text{iid}} N(0, \tau^2)$$

τ : controls the size of $\beta_2 \dots \beta_{n-1}$

$$\beta \sim N\left(\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} C & & & \\ & C & & \\ & & \tau^2 & 0 \\ & 0 & & \ddots \\ & & & & \tau^2 \end{pmatrix}\right)$$

$$\beta \sim N(0, Q)$$

$$Q = \begin{pmatrix} C & & & 0 \\ & C & & \\ & & \tau^2 & \\ 0 & & & \ddots & \\ & & & & \tau^2 \end{pmatrix}$$

likelihood: $N(X\beta, \sigma^2 I)$

prior: $N(0, Q)$

$\Downarrow$

Posterior: $\beta \mid \text{data}, \sigma^2, Q$

$$N\left(\left(\frac{X^T X}{\sigma^2} + Q^{-1}\right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + Q^{-1}\right)^{-1}\right)$$

$$Q = \begin{bmatrix} C & & & 0 \\ & C & & \\ & & \tau^2 & \\ 0 & & & \ddots & \\ & & & & \tau^2 \end{bmatrix} \quad Q^{-1} = \begin{bmatrix} 1/C & & & 0 \\ & 1/C & & \\ & & 1/\tau^2 & \\ 0 & & & \ddots & \\ & & & & 1/\tau^2 \end{bmatrix}$$

when $C \rightarrow \infty$,

$$Q^{-1} = \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & 1/\tau^2 & \\ & & & \ddots & \\ & & & & 1/\tau^2 \end{bmatrix} = \frac{1}{\tau^2} I$$

$$J = \begin{pmatrix} 0 & 0 & \dots & 0 \\ & 0 & \dots & 0 \\ & & \ddots & \vdots \\ & & & 1 \end{pmatrix}$$

Posterior Mean:

$$\begin{aligned} & \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2} \\ &= \left(\frac{X^T X}{\sigma^2} + \frac{1}{\tau^2} J \right)^{-1} \frac{X^T y}{\sigma^2} \\ &= \left(X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1} X^T y \end{aligned}$$

Recall $\hat{\beta}_{\text{ridge}}(\lambda) = (X^T X + \lambda J)^{-1} X^T y$

They are equivalent when
(Posterior Mean
A Ridge)

$$\lambda = \frac{\sigma^2}{\tau^2}$$

or $\tau = \frac{\sigma}{\sqrt{\lambda}}$

Prior:

$$\beta_0, \beta_1 \sim N(0, c) \quad \beta_2 \dots \beta_{n-1} \sim N\left(0, \frac{\sigma^2}{\lambda}\right)$$

Ridge Regression

① Least Squares $\longleftrightarrow \beta_0 \dots \beta_{n-1} \overset{i.i.d}{\sim} N(0, C)$
C large

② Ridge Regression $\longleftrightarrow \beta_0, \beta_1 \sim N(0, C),$
 $\beta_2 \dots \beta_{n-1} \sim N(0, \frac{\sigma^2}{\lambda})$

How to choose λ & σ .

$\rightarrow \beta_0, \beta_1 \sim N(0, C), \beta_2 \dots \beta_{n-1} \sim N(0, \frac{\sigma^2}{\lambda})$

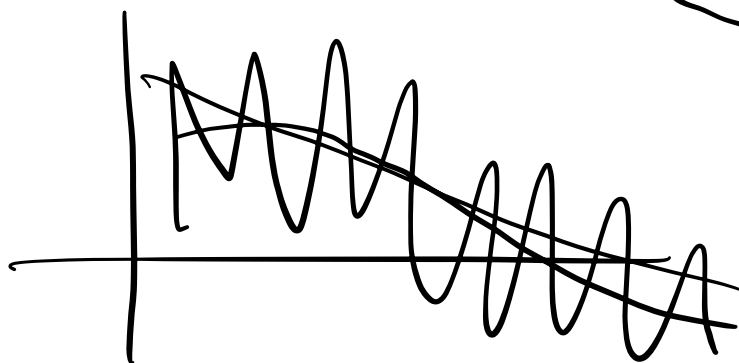
$\lambda: \lambda_{\min} \& \lambda_{\max}$

$\rightarrow \lambda \sim \text{Unif}(\lambda_{\min}, \lambda_{\max}), \log \sigma \sim \text{Unif}(C, C)$

or $\log \lambda \sim \text{Unif}(\log \lambda_{\min}, \log \lambda_{\max})$

$\rightarrow$ ① $\beta_0, \beta_1 \overset{i.i.d}{\sim} N(0, C), \beta_2 \dots \beta_{n-1} \overset{i.i.d}{\sim} N(0, C) \rightarrow$ Least Squares.

~~②~~ $\beta_0, \beta_1 \overset{i.i.d}{\sim} N(0, C), \beta_2 \dots \beta_{n-1} \overset{i.i.d}{\sim} N(0, \epsilon^2)$
for a specific small ϵ .



$$\textcircled{3} \beta_0, \beta_1 \stackrel{\text{iid}}{\sim} N(0, C), \boxed{\beta_2 \dots \beta_{n-1} \stackrel{\text{iid}}{\sim} N(0, \frac{\sigma^2}{\lambda})}$$

$$\rightarrow \log \lambda \sim \text{Unif}(\log \lambda_{\min}, \log \lambda_{\max})$$

$$\rightarrow \log \sigma \sim \text{Unif}(-C, C)$$

$$\textcircled{4} \beta_0, \beta_1 \stackrel{\text{iid}}{\sim} N(0, C), \beta_2 \dots \beta_{n-1} \stackrel{\text{iid}}{\sim} N(0, \tau^2)$$

$$\log \sigma \sim \text{Unif}(-C, C) \quad \log \tau \sim \text{Unif}(-C, C)$$

$$\beta_0, \beta_1 \stackrel{\text{iid}}{\sim} N(0, C) \quad \boxed{\beta_2 \dots \beta_{n-1} | \tau \sim N(0, \tau^2)}$$

$$\log \sigma \sim \text{Unif}(-C, C), \log \tau \sim \text{Unif}(-C, C)$$

$$\beta_0, \beta_1 \sim N(0, C)$$

$$\beta_2 \dots \beta_{n-1} | \sigma, \lambda \sim N(0, \frac{\sigma^2}{\lambda})$$

$$\log \sigma \sim \text{Unif}(-C, C)$$

$$\log \lambda \sim \text{Unif}(\log \lambda_{\min}, \log \lambda_{\max})$$

→ Likelihood.

⇒ Posterior

$$f(\beta, \sigma, \lambda | \text{data}) \propto \text{Likelihood} \times \text{prior}$$

β, σ, λ

$f(\lambda)$
 λ | data

→ after integrating
 β & σ .

